Aadesh Shah

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EDUCATION

North Carolina State University Expected December 2027 | Raleigh, NC
BS, Computer Engineering | GPA: 4.0/4.0
Relevant Coursework: Data Structures and Algorithms, Advanced Microprocessor Architecture, Parallel Architecture, Operating Systems, Linear Algebra, Embedded Systems, Digital ASIC/FPGA Design

EXPERIENCE

Software Development Engineer Intern 09/2026 – Present | Arlington, VA
Amazon Web Services (AWS)

- Developing a **full-stack agentic application** using Amazon Bedrock AgentCore and the Strands Agents SDK to generate data center simulation inputs from natural language requests, with a web UI and CI/CD pipeline.
- Built a dynamic schema and dependency-graph generator for **80+ configuration files**, enabling an AI agent to translate user intent into coordinated changes with Pydantic validation and user approval.
- Identified and fixed an S3 lifecycle configuration issue retaining incomplete multipart uploads; automated cleanup to prevent ~**$800/month** in storage costs after uncovering ~**$12K** in accumulated charges.

Software Engineering Intern 06/2026 – 08/2026 | Durham, NC
Tanium

- Built a Go pre-scan filtering service for a vulnerability scanner, excluding ~**40%** of incoming Docker images as unprocessable and reducing Lambda failure rates from **30% to 1%**.
- Engineered a Python service aggregating vulnerability scan results across **40 production targets**, integrating CloudWatch and Slack APIs to surface critical failures in real time.
- Implemented automated failure-detection and alerting infrastructure in Grafana across **10 production security services**, resolving IAM barriers to cross-region monitoring.

Applied ML Research Assistant 08/2024 – Present | Raleigh, NC
North Carolina State University

- Developed a custom CNN in **PyTorch** for respiratory signal classification from wearable sensors, achieving **98% accuracy**; applied quantization techniques for ~**4x model size reduction**.
- Owned end-to-end deployment on the MAX78000 MCU, achieving **15-16 ms latency** at ~**20 uJ/inference**; implemented full audio preprocessing pipeline in C including mel-spectrogram generation and 3-channel conversion.

R&D Embedded Software Intern 06/2025 – 08/2025 | Charlotte, NC
Columbus McKinnon

- Led C/C++ development of a next-generation control-system prototype across **5+ product lines** with direct customer input.
- Engineered task scheduling on **FreeRTOS** to improve responsiveness under strict real-time constraints.

PROJECTS

Speculative Value Prediction Engine 01/2026 – 04/2026

- Extended a large-scale C++ **RISC-V** simulator with a novel hybrid value predictor, increasing throughput by ~**25%** across 15 benchmarks within a **32 KB** budget and placing **2nd out of 60 teams** in a graduate-level competition.
- Implemented the full prediction pipeline from scratch, including training, speculative injection, and misprediction recovery, drawing on techniques from recent computer architecture research.

AI-Powered C++ Code Review & Optimization Assistant 09/2025 – 10/2025

- Built a C++ optimization pipeline using **Clang AST** analysis and an LLM feedback loop to propose source rewrites, achieving a **70%** mean runtime reduction across 22 custom benchmarks against Clang -O3 baselines.
- Developed a FastAPI/React application supporting multi-file project uploads, isolated Docker execution, optional Callgrind profiling, and optimized-source downloads with correctness checks, benchmark reports, and source diffs.

Stock Trading Application 06/2025 – 07/2025

- Developed a multithreaded C++/MySQL trading platform for **low-latency order execution and management**.
- Integrated Twitter and news APIs for market sentiment, using API caching to reduce lookup latency by ~**80%** under concurrency; fine-tuned **FinBERT** for financial sentiment classification, achieving **88% accuracy**.

SKILLS

Languages: C++, C, Go, Python, Java, Verilog, SQL
Software Engineering & Systems: Concurrency, Multithreading, CUDA, OpenMP, REST APIs, CI/CD, Linux, LLVM/Clang
Machine Learning & AI: PyTorch, BERT, CNNs, Generative AI (LLMs), Quantization
Tools & Frameworks: Git, Docker, AWS, Amazon Bedrock AgentCore, Strands Agents SDK, FastAPI, React, Vivado